COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	SECTION:	CPE21S2
DATE PERFORMED:	10-02-26	DATE SUBMITTED:	10-02-26
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ACTIVITY NO. 10 **Hands-on Activity 10.1 Graphs**

1. PROCEDURE OUTPUT

Output	PS C:\Users\TIPQC\Downloads\CPE010 - Bandigan> cd "c:\U Graph adjacency list (start_vertex, end_vertex, weight): (0, 2, 4) (0, 1, 2) (1, 4, 3) (2, 3, 2) (3, 1, 4) (4, 3, 3)

Table 10-1. Graph Output

Output	PS C:\Users\TIPQC\Downloads\CPE010 - Bandigan> cd "c:\Users\TIPQC\ 1: {2: 0}, {5: 0}, 2: {1: 0}, {5: 0}, {4: 0}, 3: {4: 0}, {7: 0}, 4: {2: 0}, {3: 0}, {5: 0}, {6: 0}, {8: 0}, 5: {1: 0}, {2: 0}, {4: 0}, {8: 0}, 6: {4: 0}, {7: 0}, {8: 0}, 7: {3: 0}, {6: 0}, 8: {4: 0}, {5: 0}, {6: 0}, DFS Order of vertices: 1 5 8 6 7 3 4 2

Table 10-2. Depth-First Search Output

Output	PS C:\Users\TIPQC\Downloads\CPE010 - Bandigan> cd "c:\Users\TIPQC\Dc 1: {2: 2}, {5: 3}, 2: {1: 2}, {5: 5}, {4: 1}, 3: {4: 2}, {7: 3}, 4: {2: 1}, {3: 2}, {5: 2}, {6: 4}, {8: 5}, 5: {1: 3}, {2: 5}, {4: 2}, {8: 3}, 6: {4: 4}, {7: 4}, {8: 1}, 7: {3: 3}, {6: 4}, 8: {4: 5}, {5: 3}, {6: 1}, BFS Order of vertices: 1 2 5 4 8 3 6 7
Observations	BFS visits vertices level by level using a queue. Starting from vertex 1, the traversal order is 1 -> 2 -> 5 -> 4 -> 8 -> 3 -> 6 -> 7, with each vertex visited once.

Table 10-3. Breadth-First Search Output

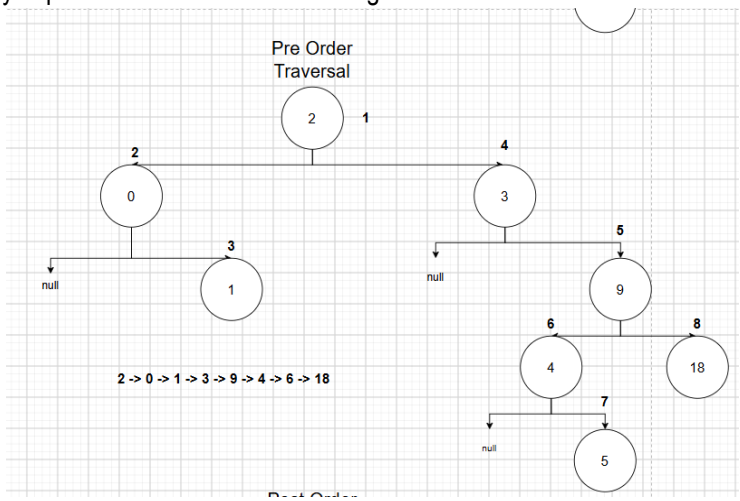
2. ANSWERS TO SUPPLEMENTAL ACTIVITY

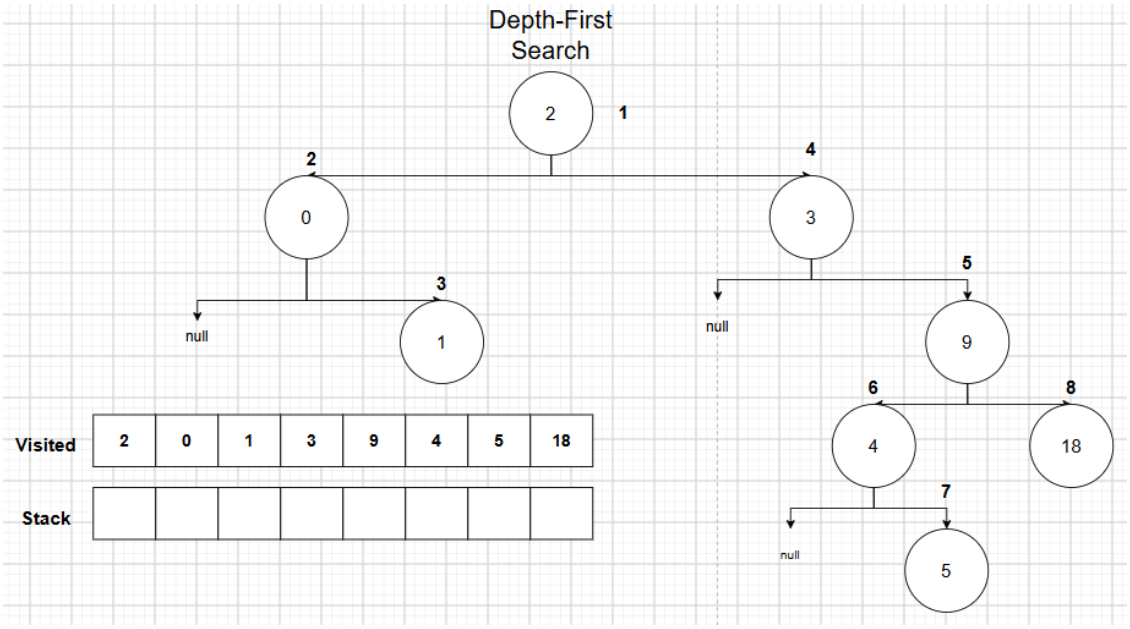
1. **A person wants to visit different locations indicated on a map. He starts from one location (vertex) and wants to visit every vertex until it finishes from one vertex, backtracks, and then explore other vertex from same vertex. Discuss which algorithm would be most helpful to accomplish this task.**

- Depth-First Search (DFS) would be the most helpful algorithm to accomplish this task because it perfectly fits what the person wants to do. A depth-first search starts from a vertex of your choice and traverses all the vertices in that branch before backtracking and visiting the other nodes.

2. **Identify the equivalent of DFS in traversal strategies for trees. To efficiently answer this question, provide a graphical comparison, examine pseudocode and code implementation.**

- The equivalent of DFS in traversal strategies for trees is the pre-order traversal. They both visit a node first and then recursively explore its children before moving to other branches.





Pseudocode

DFS
INITIALIZE stack
INITIALIZE visited list
PUSH node into stack
PUSH top of stack into visited Node
POP stack
POP adjacent nodes
REPEAT steps 3 - 7

Pre-Order
PRINT root node
IF left is not equal to null
    pre-order(left)
IF right is not equal to null
    pre-order(right)

```cpp
template <typename T>
auto depth_first_search(const Graph<T> &G, size_t dest)
{
    std::stack<size_t> stack;
    std::vector<size_t> visit_order;
    std::set<size_t> visited;
    stack.push(1); // Assume that DFS always starts from vertex ID 1
    while (!stack.empty())
    {
        auto current_vertex = stack.top();
        stack.pop();
        // If the current vertex hasn't been visited in the past
        if (visited.find(current_vertex) == visited.end())
        {
            visited.insert(current_vertex);
            visit_order.push_back(current_vertex);
            for (auto e : G.outgoing_edges(current_vertex))
            {
                // If the vertex hasn't been visited, insert it in the stack.
                if(visited.find(e.dest) == visited.end())
                {
                    stack.push(e.dest);
                }
            }
        }
    }
    return visit_order;
}
```

```cpp
template <typename T>
void BinarySearchTree<T>::preOrder(){
    std::cout << key << " ";
    if(left != nullptr){
        left->preOrder();
    }

    if(right != nullptr){
        right->preOrder();
    }
}
```

3. **In the performed code, what data structure is used to implement the Breadth First Search?**

- To implement the Breadth-First Search, the data structure queue was used. BFS starts by placing any node at the back of a queue. It then puts the front element of that queue into a visited list and adds the adjacent nodes into the queue. It repeats this until the graph is traversed.

4. **How many times can a node be visited in the BFS?**

- In a breadth-first search nodes are visited exactly once because it keeps track of which nodes are already visited.

Conclusion

In this activity I learned what graphs are and how to implement them in C++ utilizing adjacency matrix and adjacency list. I also learned two ways to traverse graphs which are the Breadth-First Search (BFS) and Depth-First Search (DFS). I was also able to implement these traversals in C++. Moreover, in the supplementary activity, I was able to answer some questions to further understand graphs.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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